

Apple iPhone Product & Market Analytics — Project Storytelling

1. Project Introduction

The objective of this project was to analyze Apple iPhone listings available on Flipkart in India and understand how **price, discounts, ratings, reviews, and product specifications** relate to customer engagement.

Instead of looking only at product prices, I wanted to answer practical business questions such as:

- Which iPhones have the highest customer ratings?
- How many customers rated the highest-rated products?
- Which iPhone has the most reviews?
- Does a higher price result in fewer customer ratings?
- Do higher discounts attract more customer ratings?
- What are the least and most expensive iPhones in the dataset?

The analysis was performed using **Python, Pandas, NumPy, Matplotlib, Seaborn, and SciPy**.

2. Dataset Understanding

The dataset contains **62 Apple product listings and 11 attributes**, including:

- Product Name
- Product URL
- Brand
- Sale Price
- MRP
- Discount Percentage
- Number of Ratings
- Number of Reviews
- UPC
- Star Rating
- RAM

The dataset is specifically based on Flipkart product listings, so the conclusions represent the products contained in this dataset rather than the entire Indian smartphone market.

3. Data Preparation

I first loaded the dataset using Pandas and inspected its structure using:

- `head()`
- `info()`
- `describe()`

The dataset contained **62 records with no missing values in the reported columns**.

I then filtered the dataset to focus specifically on products whose names contained "**iPhone**".

For the rating analysis, I converted the Star Rating field into numeric format and removed records with missing rating values before performing further analysis.

4. Finding the Highest-Rated iPhones

The first analysis focused on identifying the top 10 iPhones based on Star Rating.

The highest rating in the dataset was:

4.7 / 5

Five iPhone 11 Pro Max variants achieved this rating:

- iPhone 11 Pro Max Gold, 64 GB
- iPhone 11 Pro Max Midnight Green, 64 GB
- iPhone 11 Pro Max Space Grey, 64 GB
- iPhone 11 Pro Max Midnight Green, 256 GB
- iPhone 11 Pro Max Gold, 256 GB

Each of these variants had **1,078 ratings**.

An important observation is that the top 10 list contains product variants rather than ten completely different iPhone generations. This distinction is important when interpreting the result.

5. Customer Engagement Analysis

The project then moved beyond star ratings and examined the number of reviews.

The iPhone with the highest number of reviews was:

Apple iPhone SE (White, 256 GB)

It had:

- **8,161 reviews**
- **4.5-star rating**

This shows that the product with the largest volume of customer feedback was not necessarily the product with the highest star rating.

6. Price vs. Customer Ratings

One of the most interesting findings came from the correlation analysis between **Sale Price** and **Number of Ratings**.

The Pearson correlation coefficient was:

$$r = -0.70$$

with a reported **p-value of 0.0000**.

The scatter plot and regression line show a clear negative relationship in this dataset.

In simple terms:

Higher-priced iPhones tended to have fewer customer ratings, while lower-priced iPhones tended to have substantially more ratings.

This does not prove that price directly causes fewer ratings, but it reveals a strong association within the available data.

7. Discount vs. Customer Ratings

The second correlation analysis examined **Discount Percentage** against **Number of Ratings**.

The result was:

$$r = +0.68$$

with a reported **p-value of 0.0000**.

The visualization shows a positive relationship:

Products with higher discounts tended to receive more ratings in this dataset.

This suggests that pricing incentives may be associated with higher customer engagement, although correlation alone cannot establish causation.

8. Price Range Analysis

The project also identified the minimum and maximum priced iPhones.

Least Expensive

APPLE iPhone SE (White, 64 GB)

- Sale Price: ₹29,999
- MRP: ₹39,900
- Discount: 24%
- Ratings: 95,807
- Reviews: 8,154
- Star Rating: 4.5
- RAM: 2 GB

Most Expensive

APPLE iPhone 12 Pro (Silver, 512 GB)

- Sale Price: ₹1,40,900
- MRP: ₹1,49,900
- Discount: 6%
- Ratings: 542
- Reviews: 42
- Star Rating: 4.5
- RAM: 4 GB

This comparison provides a particularly interesting business insight: the lower-priced iPhone in the dataset had dramatically greater customer engagement than the highest-priced product.

9. Overall Business Story

The analysis tells a clear story about the products in this Flipkart dataset:

Price matters.

Lower-priced iPhones tend to have considerably more ratings, while expensive models have fewer ratings.

Discounts matter.

The positive correlation between discount percentage and ratings suggests that discounted products are associated with stronger customer engagement.

Popularity and satisfaction are different concepts.

The iPhone 11 Pro Max variants achieved the highest star rating, but the iPhone SE generated much larger volumes of ratings and reviews.

Therefore, a product with the highest rating is not necessarily the product with the highest customer engagement.

10. Technical Skills Demonstrated

This project demonstrates practical experience with:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy
- Data Cleaning
- Exploratory Data Analysis (EDA)
- Filtering and sorting
- Descriptive statistics
- Pearson correlation
- Statistical significance testing
- Data visualization
- Business insight generation

11. Final Takeaway

This project transformed raw e-commerce product data into actionable insights.

The main lesson was that **product analytics is not just about finding the highest-rated product**. A stronger analysis combines price, discount, ratings, reviews, and product specifications to understand customer behavior.

The project demonstrates how Python-based EDA can turn marketplace data into a meaningful business story and provide a foundation for more advanced analysis such as customer segmentation, price optimization, predictive modeling, or market-basket analysis.